

WJ Boshoff

Software Engineer — Java / Spring Boot · C# / .NET

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PROFESSIONAL SUMMARY

Software Engineer who delivers across the full software development lifecycle, with a focus on backend systems in the JVM and .NET ecosystems. Build scalable microservices, financial integrations, and distributed systems with Java / Spring Boot and C# / ASP.NET Core, and own them from design through production on AWS — including authentication platforms, payment integrations, data pipelines, and observability infrastructure. Equally able to work full-stack and ship cross-platform mobile apps, but strongest architecting and delivering robust server-side solutions. Currently deepening cloud and security expertise while pursuing AWS certifications.

PROFESSIONAL EXPERIENCE

Software Engineer

IdealSolution

01/2025 – Present

Centurion, Gauteng

Financial Systems & Backend Integrations

- Architected and deployed production backend integrations with multiple financial service providers including Sage (contract synchronization) and RealPay (payment processing, client onboarding, DebiCheck mandate management, and debit order automation).
- Built robust data transformation pipelines handling JSON-to-XML conversion for legacy system compatibility, ensuring reliable cross-platform financial operations.
- Engineered automated financial workflows in Spring Boot that eliminated manual processing bottlenecks and reduced error rates in client account management.

Authentication & Security Architecture

- Led the migration from AWS Cognito to a custom-built authentication service, architecting user management, role-based access control (RBAC), and multi-tenant infrastructure.
- Designed and implemented JWT-based authentication with token refresh rotation and role hierarchy management.
- Engineered row-level security (RLS) at the database layer to enforce tenant data isolation and fine-grained access control.
- Built a multi-tenant branch management system supporting complex organizational hierarchies and permission inheritance.

AI Integration & Intelligent Automation

- Integrated large language model APIs into production Spring Boot services, designing prompt strategies and structured-output contracts for reliable AI-assisted backend features.
- Built AI-powered backend components for document parsing, data extraction, and natural-language querying of business data.
- Established patterns for safe LLM integration including input validation, output schema enforcement, retry/fallback strategies, and cost/latency monitoring.

Cloud Infrastructure & DevOps (AWS)

- Owned end-to-end AWS ECS deployments including ECR image management, task definition configuration, service orchestration, and custom VPC networking.
- Implemented CI/CD pipelines using Jenkins and Azure DevOps, automating build, test, and deployment workflows.
- Configured blue-green deployments via AWS CodeDeploy, enabling zero-downtime releases and rapid rollback across three environments (ECS + Fargate).

Observability & Distributed Systems

- Architected a full-stack distributed tracing solution using OpenTelemetry, AWS Distro for OpenTelemetry (ADOT), and SigNoz on ECS.
- Implemented log aggregation and correlation across microservices, dramatically reducing mean time to resolution (MTTR) for production incidents.
- Created a data-integrity microservice in Spring Boot using dynamic IoC container configuration, enabling automated validation across distributed services, with role-based email alerting.

Enterprise Application Development

- Contributed backend development to a car rental management platform spanning lead generation, contract lifecycle, fleet tracking, and financial reporting.
- Designed CSV import/export functionality with multi-layer validation and data-quality checks, processing high-volume datasets reliably.
- Owned feature delivery from requirements through QA and production deployment; conducted code reviews and maintained high code-quality standards in an agile environment.

Mobile Contribution — World Rugby Match Tracking

- Contributed to a production Kotlin Android application used by World Rugby for live match tracking, and led a full rebuild of the companion Admin portal backend and Angular UI managing matches, tournaments, and live event data.

TECHNICAL EXPERTISE

Backend Languages & Frameworks: Java, Spring Boot (Core, Security, Data JPA), C#, .NET / ASP.NET Core, Go

Cloud & Infrastructure: AWS (ECS, Fargate, ECR, CodeDeploy, VPC, IAM, QuickSight, CloudWatch), Docker, Linux Server Administration, Azure DevOps

Databases & Data: PostgreSQL, MySQL, Oracle, SQL Optimization, Row-Level Security, Database Design, Data Migration

DevOps & CI/CD: Jenkins, GitHub Actions, Azure DevOps, Docker, Terraform, Blue-Green Deployments, Automated Testing

Observability: OpenTelemetry, AWS Distro for OpenTelemetry (ADOT), SigNoz, Distributed Tracing, Structured Logging, CloudWatch

Security & Auth: JWT, OAuth 2.0, Role-Based Access Control (RBAC), Multi-Tenant Architecture, Data Encryption

Integration & Messaging: REST APIs, ActiveMQ, Message Queues, JSON/XML Processing, Third-Party API Integration

AI & Intelligent Systems: LLM API Integration, Prompt Engineering, Structured Output Contracts, RAG Patterns

Architecture: Microservices, System Design, Design Patterns, Agile/Scrum, Git

Full-Stack & Mobile: TypeScript, Angular, React, Kotlin (Android), Dart / Flutter — full-stack and cross-platform delivery

CERTIFICATIONS & CONTINUING EDUCATION

- **Currently pursuing AWS certifications** to formalise hands-on production cloud experience across ECS, Fargate, VPC, IAM, and CodeDeploy.
- Google Cybersecurity Professional Certificate (in progress) — building on production security and authentication work.

EDUCATION

Certificate of Higher Education: Software Development

01/2023 – 11/2024

CTU Training Solutions, Potchefstroom — Graduated May 2025. Software architecture, database design, full-stack development, cloud platforms, and CI/CD automation.

Certificate of Higher Education: Networking

01/2022 – 11/2022

SELECTED PROJECTS

Local Cloud Control Plane Java 21, Spring Boot, Terraform, Docker, PostgreSQL, React

Self-built platform mimicking AWS RDS-style database provisioning on local hardware. The Spring Boot backend generates per-instance Terraform configurations and delegates container lifecycle to a Docker provider, implementing full lifecycle management, live log streaming, automatic port allocation, and encrypted credential storage. github.com/WJ-Bos/local-cloud

Yap — Voice-First Productivity App .NET 10 / ASP.NET Core Minimal API, PostgreSQL / pgvector, Whisper, Gemini 2.5 Flash, Flutter

Cross-platform app converting voice notes into structured actions. Architected the .NET 10 backend handling transcription (Whisper), structured extraction and generation (Gemini 2.5 Flash), and semantic retrieval over user history with pgvector.

Koinon — Bible Study App Flutter, Rust, PostgreSQL, LLM Ingestion & Generation

Bible study app with a Rust-backed service layer and PostgreSQL. Designed an AI ingestion-and-generation pipeline producing structured, curated study plans, with Firebase Cloud Messaging push notifications.

Boshetto — Music Theory Learning Platform TanStack Start, TypeScript, PostgreSQL, Drizzle

Music theory learning platform for guitarists with structured curriculum, ear-training, and spaced-repetition scheduling, backed by PostgreSQL via Drizzle. boshetto.com

LANGUAGES

Afrikaans — Native · English — Fluent